

Presentation Script — What to Say on Every Slide

Written in simple English. Every technical word is explained the first time it appears. Text in normal type is what you say out loud. Green text is a tip. Red text is a warning. Total speaking time: about 12 minutes.

SLIDE 1 — Where we started: the problem

About 2 minutes

Open like this

"Good morning Sir. I will explain three things. What problem we started with. What we tried. And what we actually found — which turned out to be very different from what we expected."

Explain what the system is

"We rebuilt a research paper called FedCrime. It predicts crime. But it does it in a special way. Six police stations train one shared model together. No station gives its raw data to anyone. Each station only sends the learned settings, not the actual crime records. That method is called **federated learning**. It is used when data is private."

"We ran it on real Los Angeles data from 2018. That is 113 neighbourhoods, 8 types of crime, and 365 days — one full year."

Now point at the chart

"We divided the neighbourhoods into three groups, based on how much crime they have. **Head** means the busiest areas. **Mid** is the middle. **Tail** means the quietest areas — these are mostly the poorer neighbourhoods."

"The number on each bar is the F1 score. F1 just means: out of all the times the model said "crime will happen here", how often was it right, and how many real crimes did it catch. Higher is better."

"Busy areas got **69.4**. Quiet areas got only **28.2**. The difference is **41 points**. That is a very big difference."

Then the three boxes on the right

"So it looked like the model was serving poor neighbourhoods much worse than rich ones."

"And this matters. These systems decide where police go on patrol. It is not just a number in a table. If the model fails poor areas, real people are affected."

"So our goal was simple. Close this 41-point gap. Make the model treat both kinds of area equally."

TIP: Do not rush this slide. Everything after it depends on the professor understanding this 41-point gap.

SLIDE 2 — Backup: why gated and attention are behind TCN

About 2 minutes. NOTE: this is your backup slide and it is currently in position 2.

CAREFUL: This slide is very detailed and it currently comes second. If you want a smoother story, move it to the end and use it only if the professor asks. If you keep it here, use the short version below.

First explain the words

"Before I go further, let me explain the three things we compared."

"**TCN** is the original paper's model. It only looks at time — what happened yesterday, the day before, and so on. It does not look at neighbouring areas at all. In every chart, TCN is the **dashed black line**. That is our reference point."

"**GNN** means Graph Neural Network. A graph just means we connect each area to its similar neighbours, and let them share information. The idea is: if crime is rising next door, maybe that helps predict crime here."

"We tried three kinds of graph. **Plain** is the basic one, shown in red. **Gated** is in green — it has a switch that controls how much neighbour information to take. **Attention** is in navy blue — it decides which neighbours are worth listening to."

Now the top-left chart

"These are the real scores. Look at the red bars. At 3 and 4 layers, the plain graph **completely collapses** — the score falls to 9, then almost to zero. The model basically dies."

"The green and navy bars look like they are touching the black line. But they are not. That is why I zoomed in below."

The bottom-left chart

"Same data, zoomed in. The grey shaded area is everything **below** TCN. Both gated and attention sit inside that grey area, every single time, in both cities."

"The small numbers show how far below. In Los Angeles gated is 1.04 below, then 1.04, then 0.51, then 2.11 below. In Chicago the gaps are bigger — up to 4.46 below. They never get above the line."

The right chart

"This ranks all 24 comparisons. Everything on the left of the black line is worse than TCN. **22 out of 24 are on that side**. Nine of them are big enough to be statistically real."

Now the important part — WHY

"So why does the red one collapse but the green and navy ones do not? Think of it like **mixing paint**."

"Each graph layer mixes an area with its neighbours. One layer — a little colour spreads. Fine. But the plain graph is **forced** to mix at every layer and cannot stop. After 3 or 4 layers every area is the same muddy colour. The model can no longer tell area 5 from area 40. So it cannot predict them differently. It collapses."

"The gated and attention versions have a **switch**. They learn how much neighbour information to take — anything from all of it to none of it. And what they learned was to turn it **almost off**. So they end up doing what TCN was already doing, minus a small cost."

"That is the real finding here, Sir. When the model is free to choose how much to use its neighbours, it chooses to use them almost not at all. So the problem is not which graph we picked. The neighbour information simply is not useful for this task."

TIP: SHORT VERSION if you are running out of time: "Red collapses because it cannot stop mixing. Green and navy can switch the graph off, and they do — so they land where TCN already was. Nothing beat TCN."

SLIDE 3 — How we tried to solve it

About 2 minutes

Open

"We assumed the gap was real, and we tried five different ways to fix it. Let me go through each one."

Then read each numbered point, in your own words

"**One. Reweighting.** This is the standard first thing anyone tries. You tell the model to care more about the quiet areas — give them extra importance during training. The gap went from 41 to **52**. It became **worse**, not better."

"**Two. Group-DRO.** This method looks at whichever group is doing worst, and optimises for that group. We tried three different strength settings. The gap stayed at 41. Nothing moved at all."

"**Three. Adaptive thresholds.** The model gives a probability, like 0.4 or 0.7. Then we need a cut-off to decide yes or no. Normally that cut-off is 0.5 for everything. Here we used a different cut-off for each area and each crime type. It improved a little, to about 41, then it stopped improving."

"**Four. We added a graph, a GNN.** This was your suggestion, Sir. We let neighbouring areas share information. We tested it properly — three graph types, four depths, two cities."

"**Five. Gated and attention graphs.** These are the versions specifically designed to fix the collapse problem I showed earlier. Also negative."

Then the three numbers

"Altogether: **24 tests run. 22 came out negative. Zero worked.**"

Finish with the red box — this is the turning point

"And this is the important moment. When five reasonable methods all fail, that is telling you something. It means the problem was not what we assumed it was. So we stopped trying to fix the gap, and asked instead: why does this gap exist at all?"

CAREFUL: You removed the chart from this slide. If the professor asks to see the 24 comparisons, go back to slide 2 — the ranked chart on the right side shows all of them.

SLIDE 4 — Why nothing worked: the gap is largely not real

About 2 minutes. This is the explanation slide.

Open with the main idea

"Here is what we found. The F1 score — the score everybody in this field uses — gives **free points** where crime is common."

Point at the left curve

"Imagine a model that has **zero skill**. It does not look at any data. It does not think. It just says "crime" every single day in every single place."

"In a busy area, where crime happens on about half the days, that useless model scores **66.7**. In a quiet area, where crime happens on 5 percent of days, the exact same useless model scores only

9.5."

"So that is a **57-point fairness gap** — produced by a model that never looked at the data at all."

Now connect it back

"This is why every fix failed. Most of our 41-point gap was going to be there no matter how good our model was. We were trying to close a gap that was mostly built into the **scoring system**, not into the model."

Point at the right chart

"Then we asked: is this only our problem, or is it everybody's problem? We took six published crime-prediction papers. We plotted their reported scores against how much crime each area actually has."

"**Every single line slopes upward.** Their scores follow crime volume, not model quality."

"The correlation is 0.87 — that means they move together almost perfectly. We checked four cities and found the same pattern. And the p-value is 1.3 times ten to the minus five, which means this is definitely not luck."

Then the green box — say this honestly

"One honest point, Sir. The basic mathematics behind this is already known. Davis in 2006, Chouldechova in 2017. We cite them. We do **not** claim it as ours. What is new is showing that this is actually happening across this whole field of research."

TIP: Saying this yourself is much better than a reviewer discovering it later. It shows you did your homework.

SLIDE 5 — What we found that is new

About 3 minutes. THIS IS THE MOST IMPORTANT SLIDE — slow down here.

Set it up

"Now this is the new part, and it is the strongest thing we have."

"The obvious fix is to correct for those free points — subtract them out. We built exactly that correction. And when we applied it, it said our fairness gap had closed completely. From 41 down to almost zero. A perfect result."

"It was wrong. Let me show you."

Point at the left chart, one pair of bars at a time

"This is Chicago. Grey is the model with no graph. Blue is with the graph."

"First pair — **Tail F1**, the usual fairness score for poor areas. It went **up by 12.87**. That looks like a big win."

"Second pair — **AUC**. Let me explain what AUC is. AUC asks a different question. It asks: if I show the model one day where crime happened, and one day where it did not, can it tell which is which? It does not care about the cut-off, and it does not care how rare crime is. So it **cannot be fooled**. A score of 50 means pure guessing."

"And AUC went **down by 3.05**. The model got **worse**."

"Third pair explains everything. The model went from saying 'crime here' **23 percent of the time** in poor neighbourhoods, to **80 percent of the time**. It was not finding more crime. It was just guessing more often."

Now the right chart — the decisive test

"Then we ran the test that settles the question. We took the model with **no graph at all**. No new architecture, nothing changed inside it. We only moved the decision cut-off, that one number that decides yes or no."

"The green dashed line is AUC. It is completely flat at 61.99, on all nineteen rows. It has to be flat, because AUC does not depend on the cut-off at all."

"The blue line is F1, and it moves a lot. And at cut-off 0.35 — that is the orange star — it reaches **41.59**. Which is **better** than what the graph model achieved."

Then the four number boxes

"So: the graph model, with all that extra machinery, gave us plus 12.87. Just moving one number gave us plus **13.63** — bigger. AUC stayed identical throughout. And the model went from crying "crime" 23 percent of the time to 80 percent."

Finish with the red box

"The model did not get better at finding crime. It started labelling 8 out of 10 poor neighbourhoods as high-crime. And the standard fairness score **rewarded** it for doing that."

"If a police department used this, their fairness dashboard would show a big improvement. Meanwhile the system would be less accurate, and sending more patrols into poor areas."

CAREFUL: Check the box that says "80% → 23%". It should read "23% → 80%" — it starts at 23 and goes UP to 80. The arrow may have flipped when you edited the file.

SLIDE 6 — What is new, and where we can publish it

About 2 minutes. Finish strong, then ask your question.

The three contributions

"So what can we actually claim as ours? Three things."

"**Number one, and this is the strongest.** Correcting for base rates is **not enough**. A model can fake a fairness win just by guessing more often. Our own correction was completely fooled by this. Only AUC caught it. As far as we can find, nobody has stated this clearly before."

"**Number two.** Published fairness gaps in this field follow crime volume. Six papers, correlation 0.87 to 0.90. And it holds even when we use no statistical assumptions at all — just the ordering of the groups, which the papers state directly."

"**Number three.** It changes who gets help."

Point at the two circles

"Choosing which **model** is best — completely unaffected. Zero out of five changed. That is good news, and we report it honestly."

"But choosing which **area** needs extra resources — that flipped for **every single model**. Six out of six."

"So the flaw does not mislead us about which method to use. It misleads us about **where** the model is failing. And that is the whole fairness question."

The publication plan

"For publication I want to target **ACM FAccT 2027**. That is the ACM Conference on Fairness, Accountability and Transparency. It is the main conference in the world for this exact topic."

"The abstract is due on 27 October, the full paper on 3 November. That gives us about eleven weeks."

"Acceptance is about 27 percent. But they also have a revision round — papers sent there get accepted 71 percent of the time. So the real path is: get a revise decision, fix what they ask, and it very likely goes through."

"And their call for papers asks specifically for work that 'examines evaluation metrics and measurement practices'. That is our paper, described in their own words."

"If FAccT does not work out, the backup is ACM TIST, which is a journal with no deadline."

End with your question — do not end with a conclusion

"So the decision I need from you, Sir. Should we build the paper around this measurement finding, and keep the attack and defence work for a second paper? The attack results were produced before we fixed a reproducibility problem, so those numbers would need re-running anyway."

"Thank you. I am happy to take any questions."

If he asks these questions

Short answers you can give straight away.

Question	Your answer
Did the gated GNN work?	"No Sir. It was negative on both cities, at every depth. Slide 2 shows it. That is why I do not think it can be the paper's main contribution."
Is this not already known?	"The basic formula is, and we cite it. What is new is that correcting for it is not enough, and that we can show a model faking a fairness win."
How reliable are these numbers?	"Every experiment is five separate runs with fixed random seeds. We found and fixed a bug where the same command gave different answers. Now it is identical to six decimal places on the same machine."
Why is the graph still in the paper?	"It is how we discovered the problem. The threshold experiment is the general result. The graph is the case study that led us there."
What is AUC in one line?	"Can the model tell a crime day from a no-crime day? 50 means pure guessing. It ignores the cut-off, so you cannot cheat it."
Why did you not just use a better model?	"We tried. Three graph types, four depths, two cities. The best ones learned to switch themselves off. The information is not there to be used."
What is left to do?	"All the experiments are finished. What remains is writing the paper. About four to six weeks."

Three rules for the day

1. Never say the gated GNN worked better than TCN. It did not. All the bars are below the line. Say instead: "it did not damage the model, but it did not beat TCN either — it learned to switch itself off."
2. Say the $2p/(1+p)$ formula is known and cited, before he asks. It protects you.
3. End on the question, not a conclusion. You want a decision from him, not applause.